

Adaptive Civic Routing Intelligence System

LIVE MVP DEPLOYED

ARCHITECTURAL DECISION RECORD (ADR) - HACKMYSURU 1.0 - ROUTING

Team Spark | ID: HM26-E9C5 | Single-Page Architectural Ledger

adaptive-civic-routing.vercel.app

Core Principle: AI assists issue categorization; PostGIS and relational rules strictly decide legal jurisdiction & department.

DECISION Q1

Deterministic PostGIS Containment vs. Probabilistic Routing

[CHOSEN DECISION]

PostgreSQL 18 + PostGIS 3.6 spatial geometry engine. Complaints store SRID 4326 Point coordinates. Authority resolution executes deterministic ST_Covers against active ward MultiPolygons.

[REJECTED ALTERNATIVES]

1. Probabilistic LLM/NLP classification from complaint text (hallucinates on border zones). 2. Radial bounding boxes (wards are non-convex). 3. Reverse geocoding APIs (lack municipal authority bounds).

Why Approach Was Selected:

Municipal authority (MCC vs. MUDA) is governed by statutory legal boundaries. PostGIS geometric evaluation is 100% mathematically reproducible, eliminates hallucinations, and creates legally defensible civic audits.

Main Trade-off & Mitigation:

Requires maintaining verified GIS shapefiles in database. Coordinates outside mapped boundaries fail auto-routing and quarantine safely into the HUMAN_REVIEW queue for operator triage.

DECISION Q2

Immutable Jurisdiction Versioning vs. Mutable Boundary Overwrites

[CHOSEN DECISION]

Decouple ward boundaries into explicit lifecycle states (DRAFT -> ACTIVE -> RETIRED). New tickets route against MYS_2026_V2 (ACTIVE); past tickets remain locked to MYS_2026_V1 (RETIRED) via foreign key in routing_decisions.

[REJECTED ALTERNATIVES]

1. In-place polygon updates overwriting geometry coordinates directly. 2. Dynamically re-evaluating historical tickets against current boundaries. 3. Unversioned GIS layers lacking delimitation tracking.

Why Approach Was Selected:

Mysuru ward boundaries periodically delimit (e.g. MCC expanding into MUDA sectors). MYS_2026_V2 is active; past tickets under MYS_2026_V1 remain mathematically true to the day of routing, ensuring legal municipal provenance.

Main Trade-off & Mitigation:

Additional relational schema complexity and storage overhead to retain retired geometries. Mitigated by atomic version transition transactions (deactivate old, activate new in one BEGIN...COMMIT block).

DECISION Q3

In-Database Spatial Routing & Municipal Scaling Architecture

[CHOSEN DECISION]

Execute point-in-polygon containment synchronously inside PostgreSQL at complaint intake. Guarantees sub-50ms latency for Mysuru's 65 wards and ensures immediate transactional consistency and receipt generation.

[REJECTED ALTERNATIVES]

1. Client-side Turf.js evaluation (leaks boundary data, vulnerable to client tampering). 2. External GIS SaaS APIs (introduces third-party per-request billing, internet failure points, breaks ACID transactions).

Why Approach Was Selected:

In-database execution ensures ACID consistency at ticket creation (POST /api/complaints), eliminates external API failure points, and guarantees atomic binding without third-party dependencies.

Main Trade-off & Mitigation:

Concurrent spatial queries during civic storm events can burden database CPU. Roadmap: (1) Bounding-box pre-filter (&&); (2) Range table partitioning by fiscal year/version; (3) BullMQ async queue for peak spikes.